

Task 2
Retail Customer Retention Analytics – TARGET



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Tool Used: **POWER BI**

Video Explanation link:
<https://www.loom.com/share/d6f3272e620047a793b862b13c637dac>

Abstract / Executive Summary

This project focuses on analysing Target Retail customer data to understand customer retention, churn behaviour, loyalty engagement, repeat purchases, and customer lifetime value using Microsoft Power BI.

Multiple datasets including customer demographics, transaction history, loyalty information, store details, and churn labels were integrated to create an interactive dashboard that provides business insights. The dashboard helps identify high-value customers, churn patterns, loyalty performance, and store-wise retention trends, enabling better decision-making for customer retention strategies.

Objectives

1. Analyze customer churn and retention using Power BI.
2. Identify repeat purchase behavior across customer segments.
3. Evaluate Customer Lifetime Value (CLV).
4. Measure promotion and loyalty program effectiveness.
5. Compare store and channel performance.
6. Provide recommendations to improve customer retention.

Tools & Technologies Used

Tool	: Purpose
Microsoft Power BI	: Dashboard Creation
Power Query	: Data Cleaning
DAX	: Measures & Calculations
CSV Files	: Data Source
Data Modeling	: Table Relationships
KPI Cards & Charts	: Visualization

Dataset Description

Customer Demographics

Contains Customer_ID, Age, Gender, Region, Income_Level, Membership_Since, Preferred_Channel.

Customer Transactions

Contains Transaction_ID, Customer_ID, Store_ID, Product Category, Transaction Date, Amount and Promotion Applied.

Store Locations

Contains Store_ID, Store Type, Region and Opening Year.

Loyalty Program

Contains Loyalty Tier, Points Earned and Points Redeemed.

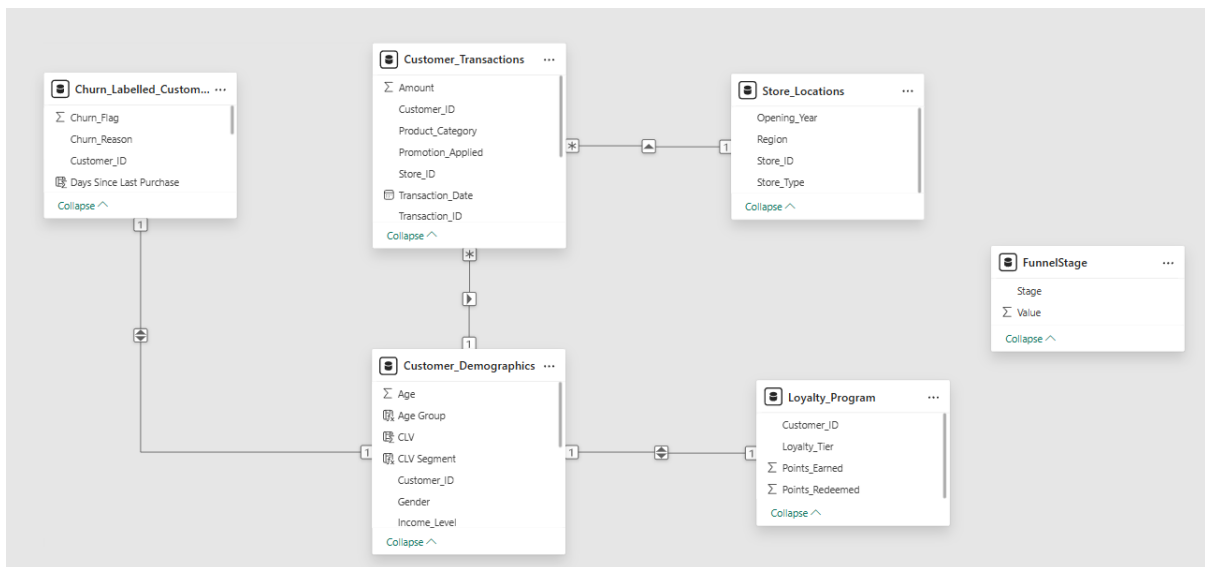
Churn Customers

Contains Customer_ID, Last Purchase Date, Churn Flag and Churn Reason.

Methodology / Project Workflow

Task 1: Data Cleaning & Modeling

- Imported all datasets into Power BI.
- Removed duplicate records and handled missing values.
- Converted data types appropriately.
- Created calculated columns:
 - Membership Duration
 - Transaction Year
 - Transaction Month
- Built relationships between all tables using Customer_ID and Store_ID.



DAX used:

Membership Duration (Calculated Column)

Membership Duration =

```

DATEDIFF(
    Customer_Demographics[Membership_Since],
    TODAY(),
    YEAR
)

```

Purpose: Calculates how many years the customer has been a member.

2. Transaction Year

Transaction Year =

```

YEAR(Customer_Transactions[Transaction_Date])

```

Purpose: Extracts the transaction year.

3. Transaction Month

Transaction Month =

```

FORMAT(Customer_Transactions[Transaction_Date],"MMMM")

```

Purpose: Extracts the month name.

Task 2: Churn & Retention Metrics

Created KPI Measures:

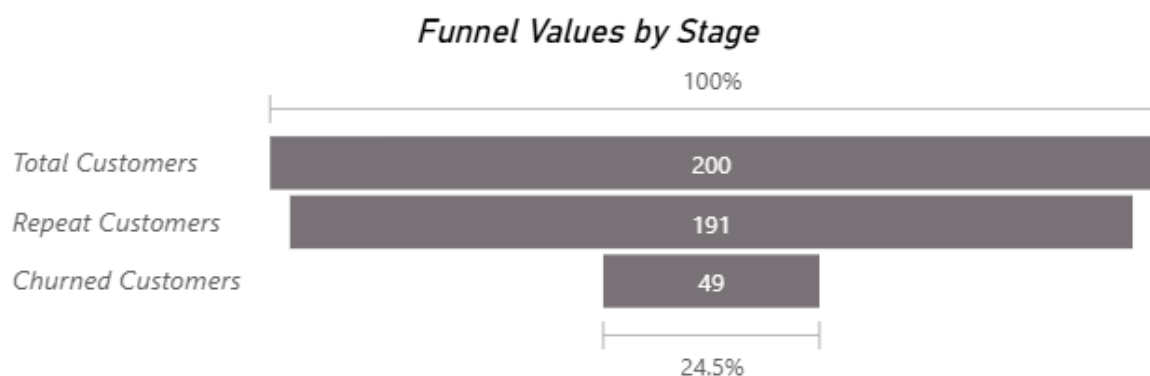
- Total Customers
- Churned Customers
- Repeat Customers
- Churn Rate %

Created visuals for:

- Churn by Region
- Churn by Income Level
- Churn by Preferred Channel
- Churn by Loyalty Tier

Used Funnel Chart for:

Total Customers → Repeat Customers → Churned Customers



DAX USED:

1. Total Customers

Total Customers =

DISTINCTCOUNT(Customer_Demographics[Customer_ID])

2. Churned Customers

Churned Customers =

```
CALCULATE(  
    COUNT(Churn_Labelled_Customers[Customer_ID]),  
    Churn_Labelled_Customers[Churn_Flag]=1  
)
```

3. Churn Rate

```
Churn Rate =  
DIVIDE(  
    [Churned Customers],  
    [Total Customers],  
    0  
) * 100  
4. Repeat Customers  
Repeat Customers =  
CALCULATE(  
    DISTINCTCOUNT(Customer_Transactions[Customer_ID]),  
    FILTER(  
        VALUES(Customer_Transactions[Customer_ID]),  
        CALCULATE(COUNT(Customer_Transactions[Transaction_ID]))>1  
    )  
)
```

```
5. Funnel Values  
Funnel Values =  
SWITCH(  
    SELECTEDVALUE(FunnelStage[Stage]),  
    "Total Customers",[Total Customers],  
    "Repeat Customers",[Repeat Customers],  
    "Churned Customers",[Churned Customers]  
)
```



Task 3: Repeat Purchase Analysis

Created:

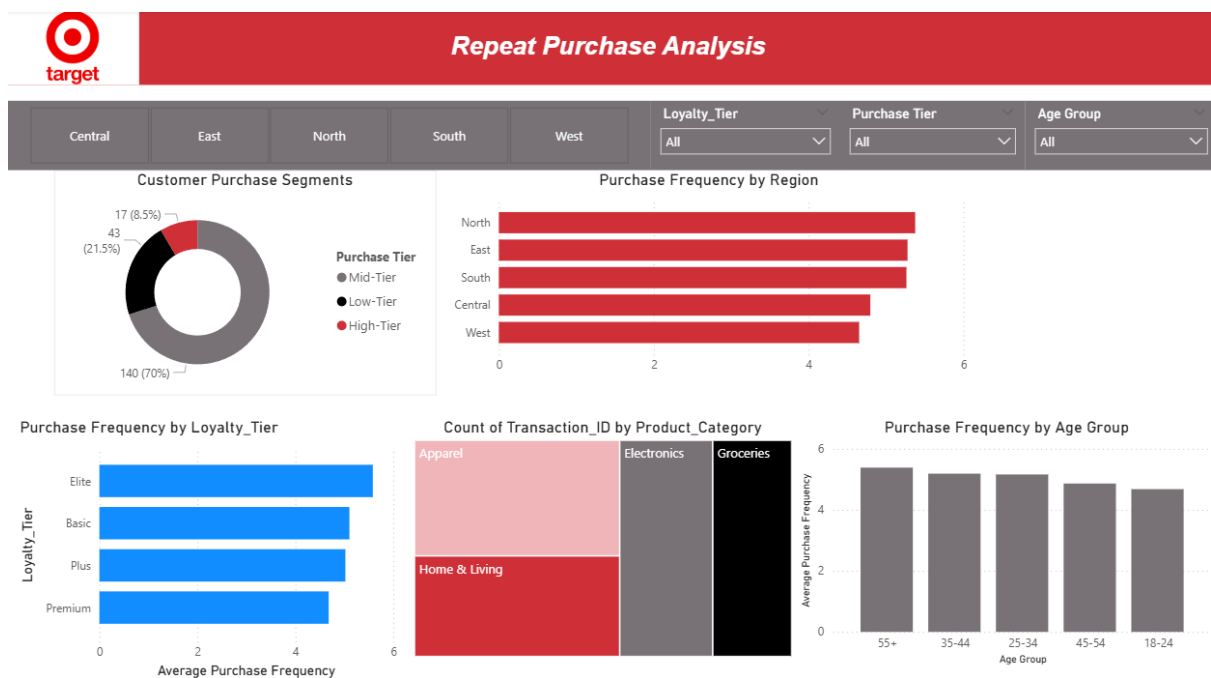
- Repeat Rate
- Average Purchases
- Repeat Customer Segments

Segments:

- Low Tier
- Mid Tier
- High Tier

Visualizations:

- Avg Purchases by Region
- Avg Purchases by Loyalty Tier
- Avg Purchases by Age Group
- Most Purchased Categories
- Repeat Segment Distribution



DAX USED:

1.Purchase Count

Purchase Count =

```
CALCULATE(  
    COUNT(Customer_Transactions[Transaction_ID])  
)
```

2. Purchase Tier

Purchase Tier =

```
SWITCH(  
    TRUE(),  
    [Purchase Count]<=3,"Low-Tier",  
    [Purchase Count]<=8,"Mid-Tier",  
    "High-Tier"  
)
```

3. Age Group

Age Group =

```
SWITCH(  
    TRUE(),  
    Customer_Demographics[Age]<25,"18-24",  
    Customer_Demographics[Age]<35,"25-34",  
    Customer_Demographics[Age]<45,"35-44",  
    Customer_Demographics[Age]<55,"45-54",  
    "55+"  
)
```

4. Average Purchase Frequency

Average Purchase Frequency =

```
AVERAGE(Customer_Demographics[Purchase Count])
```


Task 4: Promotion & Loyalty Impact

Calculated:

- Promotion Transaction %
- Average Amount with Promotion
- Average Amount without Promotion

Compared:

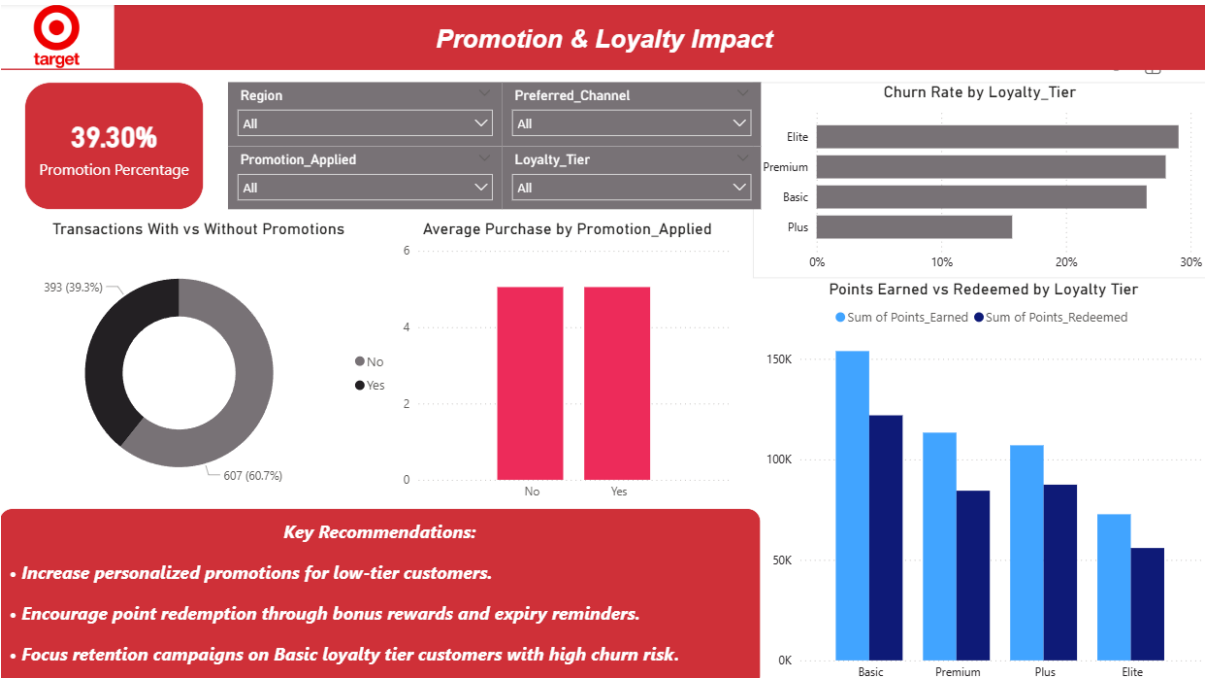
- Loyalty Tier Churn
- Points Earned vs Redeemed

Visuals:

- Donut Chart
- Clustered Column Chart
- KPI Cards

Recommendations:

- Improve loyalty rewards
- Increase personalized offers
- Enhance customer engagement



DAX USED:

1. Promotion Percentage

Promotion Percentage =

```
DIVIDE(  
CALCULATE(  
COUNT(Customer_Transactions[Transaction_ID]),  
Customer_Transactions[Promotion_Applied]="Yes"  
),
```

```
COUNT(Customer_Transactions[Transaction_ID]),  
0  
)*100
```

2. Average Purchase Amount

Average Purchase Amount =

```
AVERAGE(Customer_Transactions[Amount])
```

Task 5: Store & Channel Performance

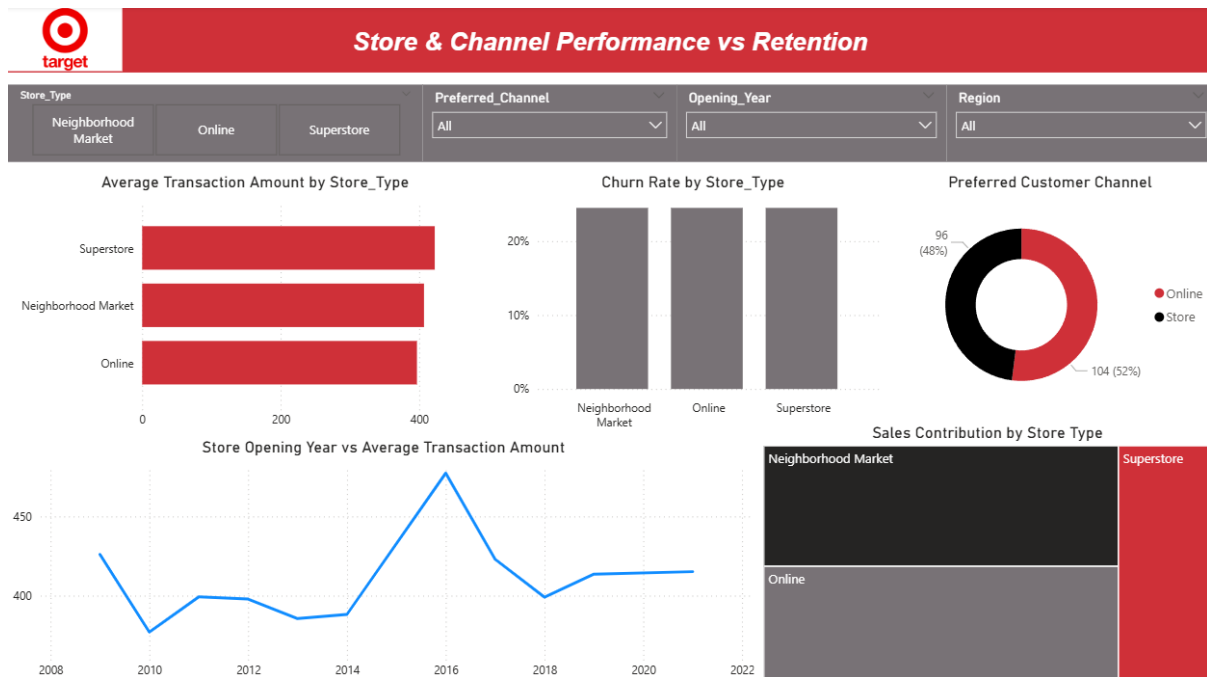
Merged Store and Transaction datasets.

Analyzed:

- Average Transaction Amount by Store Type
- Churn Rate by Store Type
- Correlation between Store Opening Year and Retention

Visuals Used:

- Bar Chart
- Scatter Plot



DAX USED:

1. Average Transaction Amount

Average Transaction Amount =
 $AVERAGE(Customer_Transactions[Amount])$

2. Retention Rate

Retention Rate =
 $100 - [Churn Rate]$

Task 6: Customer Lifetime Value (CLV)

Calculated:

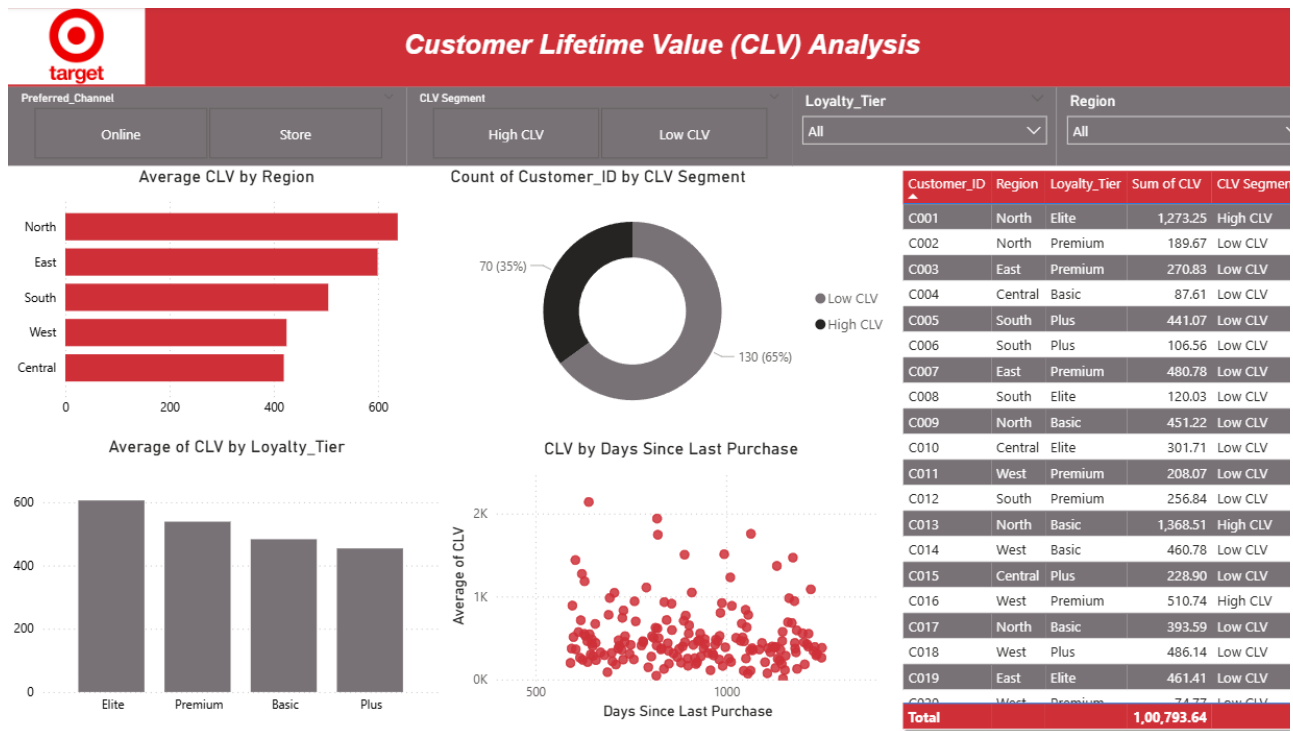
$CLV = Total\ Amount\ Spent / Membership\ Duration$

Created:

- High CLV
- Low CLV Segments

Visualizations:

- CLV vs Days Since Last Purchase
- CLV by Region
- CLV by Loyalty Tier



DAX USED:

1. Total Amount Spent

Total Amount Spent =

```
CALCULATE(
SUM(Customer_Transactions[Amount])
)
```

2. CLV

CLV =

```
DIVIDE(
Customer_Demographics[Total Amount Spent],
Customer_Demographics[Membership Duration],
0
)
```

3. Average CLV

Average CLV =

```
AVERAGE(Customer_Demographics[CLV])
```

4. CLV Segment

CLV Segment =

VAR AvgCLV =

```
CALCULATE(
```

```
AVERAGE(Customer_Demographics[CLV]),  
ALL(Customer_Demographics)  
)  
RETURN  
IF(  
Customer_Demographics[CLV]>AvgCLV,  
"High CLV",  
"Low CLV"  
)
```

5. Days Since Last Purchase

```
Days Since Last Purchase =  
DATEDIFF(  
Churn_Labelled_Customers[Last_Purchase_Date],  
TODAY(),  
DAY  
)
```

Dashboard Overview

The dashboard combines all important customer retention insights into multiple interactive pages.

Main KPIs include:

- Total Customers
- Churn Rate
- Repeat Rate
- CLV
- Repeat Customers
- Churn Customers

Interactive slicers:

- Region
- Income Level
- Channel
- Loyalty Tier
- Year



Target Retail Customer Retention Analytics

Churn & Retention Metrics

200

Total Customers

24.50%

Churn Rate

406.07K

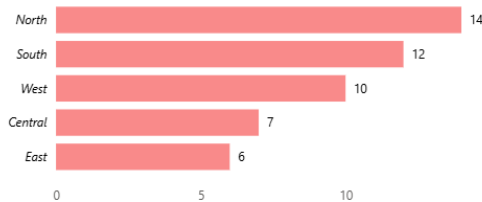
Total Sales

191

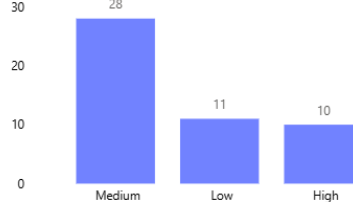
Repeat Customers

Region	Preferred_Channel
All	All
Income_Level	Loyalty_Tier
All	All

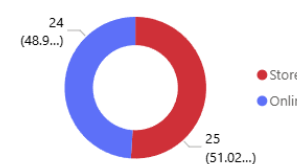
Churn by Region



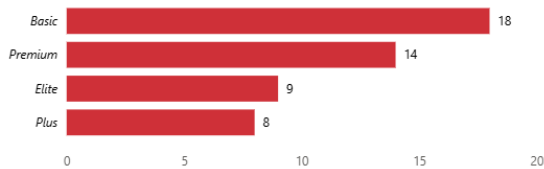
Churn by Income Group



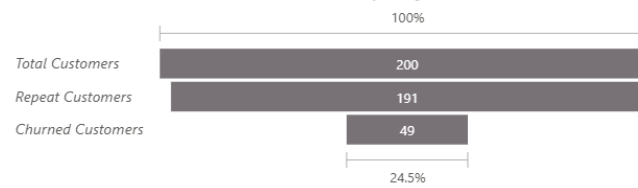
Churn by Channel



Churn by Loyalty Tier



Funnel Values by Stage



Key Insights

- Customer churn varies across different regions and loyalty tiers.
- High loyalty customers demonstrate better retention.
- Promotion campaigns positively impact transaction frequency.
- Repeat customers contribute significantly to revenue.
- High CLV customers should be prioritized for retention strategies.

Top 3 Recommendations

1. Which customers should be prioritized for retention?

Recommendation:

Target should prioritize high-value customers, including those with high Customer Lifetime Value (CLV), Premium and Elite loyalty memberships, and repeat purchase behaviour. These customers contribute significantly to the company's revenue and profitability. Personalized offers, exclusive discounts, and loyalty rewards should be provided to maintain their engagement and reduce the risk of churn.

2. Which channels are underperforming?

Recommendation:

The analysis indicates that the channel with lower repeat purchases and higher churn rates is underperforming. Target should improve customer experience across this channel by enhancing service quality, introducing targeted promotions, and strengthening the integration between online and offline shopping experiences. These improvements can help increase customer satisfaction and retention.

3. How can loyalty engagement be strengthened?**Recommendation:**

Target can strengthen loyalty engagement by offering personalized rewards, bonus loyalty points, exclusive member benefits, and timely reminders for point redemption. Regular promotional campaigns and customized offers based on customer purchase history can encourage repeat purchases, increase loyalty program participation, and improve long-term customer retention.

These three recommendations are concise, professional, and directly address the questions asked in the project, making them ideal for inclusion in your Executive Summary or Final Dashboard.

Conclusion

The Target Retail Customer Retention Analytics Dashboard provides a comprehensive view of customer behavior, loyalty performance, churn trends, and purchase patterns. Using Power BI, business users can monitor KPIs in real time and make data-driven decisions to improve customer satisfaction and long-term retention.

References

- Walmart Retail Customer Data
- Microsoft Power BI Documentation
- DAX Guide (SQLBI.com)